



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

RI2002-1 – DESIGN OF ROBOTICS COMPONENTS

(RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

❖ Use of the Design Data handbook is permitted.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is not a reason to design and redesign a product?
 A) Optimum design
 B) Innovation
 C) Appearance
 D) None of the above
- In the design process, which step is followed after defining the problem?
 A) Analysis
 B) Synthesis
 C) Optimization
 D) Evaluation
- Which of the following factors are not considered while selecting values for a factor of safety?
 A) Cost
 B) Failure Consequences
 C) Degree of uncertainty about loading and material
 D) None of the above
- What is the expression for the modulus of rigidity in terms of the modulus of elasticity and the Poisson ratio?
 A) $G = 3E / 2(1 + \mu)$
 B) $G = 5E / (1 + \mu)$
 C) $G = E / 2(1 + \mu)$
 D) $G = E / (1 + 2\mu)$
- E, G, K and μ elastic modulus, shear modulus, bulk modulus, and Poisson's ratio respectively. To express the stress-strain relations completely for this material, at least _____
 A) E, G, and μ must be known
 B) E, K, and μ must be known
 C) Any two of the four must be known
 D) All the four must be known
- In a tensile test, near the elastic limit zone _____
 A) Tensile stress increases in linear proportion to the stress
 B) Tensile stress increases at a faster rate
 C) Tensile stress decreases at a faster rate
 D) None of the mentioned
- For an elliptical hole on a flat plate, if the width of the hole in the direction of the load decreases, the Stress Concentration Factor will _____
 A) Increase
 B) Decrease
 C) Remains constant
 D) Can't be determined. Varies from material to material
- In which of the following cases stress concentration factor is ignored?
 A) Ductile material under static load
 B) Ductile material under fluctuating load
 C) Brittle material under static load
 D) Brittle material under fluctuating load
- Which of the following requires more space for the rotation of the spanner?
 A) Square Head
 B) Hexagonal Head
 C) Both require equal space of rotation
 D) Cannot be stated
- Set screws are subjected to tensile forces only.
 A) Yes
 B) No, they are subjected to compressive forces only
 C) Both compressive and tensile
 D) Can't be determined
- The lead of a single start thread is _____ its pitch
 A) double then
 B) triple then
 C) equal to
 D) half then

12. When two pulleys of different diameters are connected using an open belt drive, then the angle of contact taken into consideration should be the
- A) larger pulley
B) smaller pulley
C) average of two pulleys
D) none of the mentioned
13. The centrifugal tension in belts
- A) increases power transmitted
B) decreases power transmitted
C) have no effect on the power transmitted
D) increases power transmitted up to a certain speed and then decreases
14. When the belt is stationary, it is subjected to some tension, known as initial tension. The value of this tension is equal to the
- A) tension in the tight side of the belt
B) tension in the slack side of the belt
C) sum of the tensions in the tight side and slack side of the belt
D) average tension of the tight side and slack side of the belt
15. Which of the following cannot take radial load?
- A) Cylindrical Roller bearing
B) Taper roller bearing
C) Thrust ball bearing
D) None of the listed
16. In angular contact bearings, ____ bearings are required to take thrust load in both directions.
- A) 1
B) 4
C) 2
D) 3
17. Is it necessary for an axle to be _____ concerning a rotating element?
- A) Stationary
B) Moving
C) Moving or stationary
D) None of the listed
18. Which of the following act on shafts?
- A) Torsional moment
B) Bending Moment
C) Both torsional and bending
D) None of the mentioned
19. Which of the following is true for worm gears?
- A) The worm is in the shape of a threaded screw
B) Threads on the worm have a small lead
C) Worm imposes high thrust loads
D) Characterized by low-speed reduction ratio
20. The greater the velocity ratio, the smaller the gearbox.
- A) True
B) Greater the gearbox
C) The size of gearbox remains unaffected
D) None of the listed

PART - B: DESCRIPTIVE ANSWER QUESTIONS.

Unit – I

1. a) A 50mm diameter steel rod supports a 9kN load and in addition, is subjected to a torsional moment of 100N-m as shown in Fig Q1(a). Determine the maximum tensile and the maximum shear stress.
- b) Determine the maximum principal stress and maximum shear stress at sections A-A of the member shown in Fig Q1(b). Neglect transverse shear.

Marks BT* CO* PO*

08 L2 1 1

08 L2 1 1

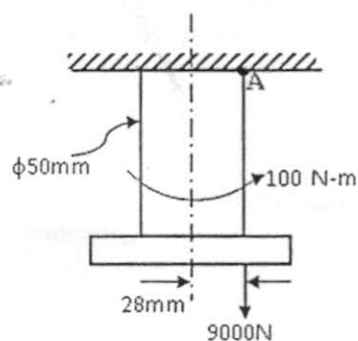


Fig Q1(a)

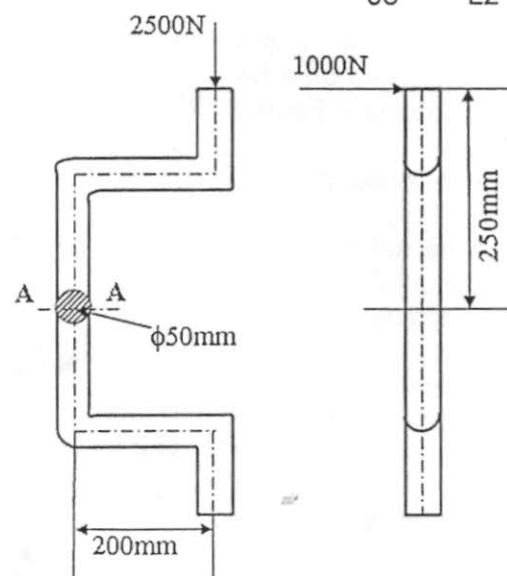


Fig Q1(b)

2. a) The following data refers to a mild steel specimen tested in a laboratory.

Diameter of specimen = 25mm

Extension under a load of 20kN = 0.04mm

Maximum load = 225 kN

Neck diameter = 18.25 mm

The gauge length of the specimen = 200 mm

Load at yield point = 150kN

Length of the specimen after failure = 275 mm

Determine: (i) Young's Modulus, (ii) Yield Stress
(iii) Ultimate Stress (iv) Percentage Elongation
(v) Percentage Reduction in Area
(vi) Safe Stress Adopting a Factor of Safety of 2.5.

12 L2 1 1

- b) A 25mm diameter rod was subjected to a tensile load of 50 kN. The extension of the rod was found to be 250 divisions in the extensometer of 200mm gauge length. If each division of the extension meter is equal to 0.001mm, find the modulus of Elasticity.

04 L3 1 1

3. a) Find the diameter of the hole shown in **Fig Q3(a)**, if the stress concentration factor at the hole is to be the same as the fillet.

08 L2 1 1

- b) A grooved shaft in **Fig Q3(b)**, is to transmit 5kW at 120rpm. Determine the diameter of the shaft at the groove if it is made of C15 steel. A factor of safety is 2.

08 L3 1 1

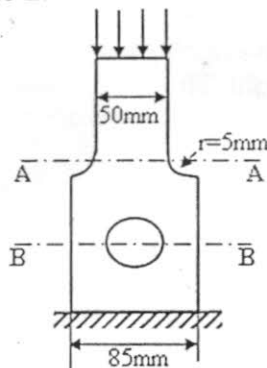


Fig Q3(a)

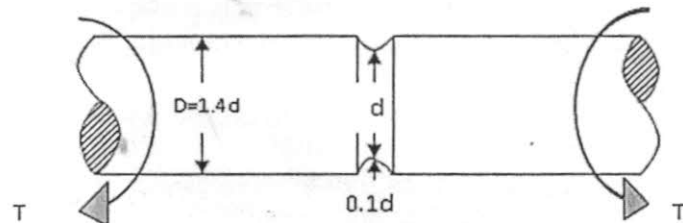


Fig Q3(b)

Unit – II

4. a) A pillar crane is fastened to the foundation by 12 bolts spaced equally on a bolt circle of 1.8 m diameter as shown in **Fig Q4(a)**. Determine the size of the bolt when a load of 40 kN acts at a radius of 8 m. The Diameter of the pillar flange is 2.1 m. The allowable stress intensity for the bolt material is 90 MPa.

06 L2 2 1

- b) The structural connection shown in **Fig Q4(b)** is subjected to an eccentric load P of 10kN with an eccentricity of 500 mm. The center distance between bolts at 1 and 3 is 150 mm and the center distance between bolts at 1 and 2 is 200 mm. All bolts are identical. The bolts are made of plain carbon steel having yield strength in tension of 400MPa and the factor of safety is 2.5. Determine the size of the bolts.

10 L3 2 1

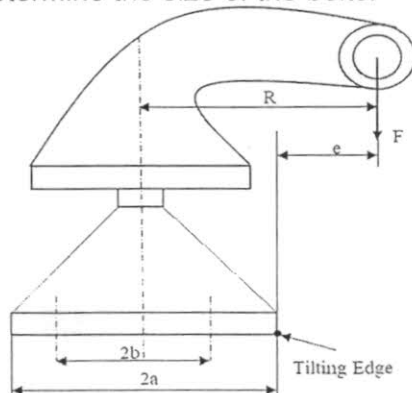


Fig Q4(a)

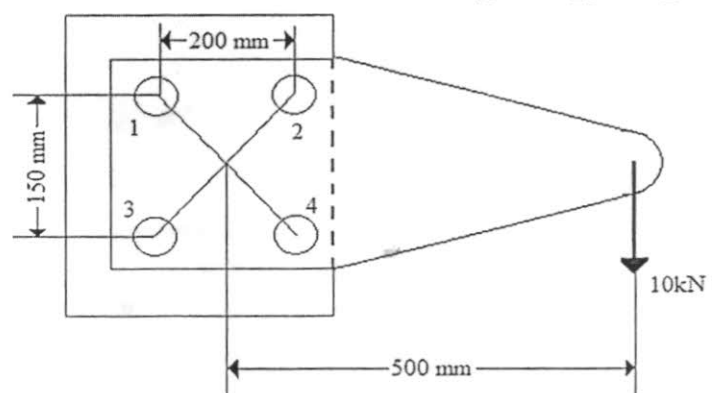


Fig Q4(b)

5. A V-belt is to be arranged between two shafts whose centers are 3000 mm. The driving pulley is of 850 mm effective diameter and is to be supplied with 75 kW at 960 rpm. The follower pulley is to run at 480 rpm. Determine the number of belts required for the following particulars.
 Area of belt section = 400 mm^2
 Weight of belt = 0.01 N/cm^3
 Safe working tensile stress = 2.1 N/mm^2
 Coefficient of friction = 0.27
 Groove angle of pulley = 40°
 Also, find the initial tension required in each belt. 16 L4 2 2
6. a) A single-row deep groove ball bearing has a specific dynamic capacity of 46.3 kN. The actual radial load $F_r = 9 \text{ kN}$. The speed of rotation is 1800 rpm. What is the life in i) Cycle of operation ii) In hours, iii) What is the average life? 10 L1 2 1
- b) A taper roller bearing has a dynamic load capacity of 26 kN. The desired life for 90% of the bearings is 8000h and the speed is 300 rpm. Calculate the equivalent radial load that the bearing can carry. 06 L2 2 1

Unit – III

7. A shaft is supported by two bearings placed 1100mm apart. A pulley of diameter 620mm is keyed at 400mm to the right from the left-hand bearing and this drives a pulley directly below it with a maximum tension of 2.75kN. Another pulley of diameter 400mm is placed 200mm to the left of the right-hand bearing and is driven with a motor placed horizontally to the right. The angle of contact of the pulleys is 180° and $\mu = 0.3$. Determine:
 i. The diameter of the shaft.
 ii. Draw the bending moment diagram.
 Assume $C_m = 3$, $C_t = 2.5$, and $\tau = 40.5 \text{ MPa}$. 16 L3 3 2
8. a) Classify the different types of gears. 04 L2 3 1
- b) A pair of spur gears consists of a 20 teeth pinion meshing with a 120 teeth gear. The module is 4mm. Calculate (i) The center distance (ii) the pitch circle diameters of the pinion and the gear (iii) The addendum and dedendum (iv) The tooth thickness (v) The bottom clearance and (vi) The gear ratio. 12 L3 3 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
